

描述性分析

时序图. ACF图. $X_t = f(T_t, S_t, I_t)$

trend 趋势
season 季节

一. 趋势平滑 trend smoothing

1. 不含季节性

(1) 参数回归法 线性: $X_t = \beta_0 + \beta_1 t + I_t$

非线性: (Logistic 函数): $X_t = \frac{a}{1 + be^{-ct}} + I_t$

缺陷: 定义的全局参数不能代表序列的真正趋势.

(2) 移动平均法 $\hat{T}_t = \frac{X_{t-h} + \dots + X_{t-1} + X_t + X_{t+1} + \dots + X_{t+h}}{2h+1}$ (Moving Average)

在 t 左右各取 h 个点 (记作 h -MA), 想法: 通过取平均来消除波动.

$\Rightarrow$ [推广] 线性滤波. 权重 $w_i = (\frac{1}{2} + \frac{1}{2}s)^i = \frac{1}{2^{2h}} + (1+s)^h$, $\sum w_i = 1$.

例: $h=1$ 时, $\hat{T}_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$ (无法估计两端)

(3) 指数平滑法 (解决对称滤波的问题 $\Rightarrow$ 非对称) (ETS)

$$X_t = \alpha X_t + \alpha(1-\alpha)X_{t-1} + \alpha(1-\alpha)^2 X_{t-2} + \dots = \sum_{j=0}^{\infty} \alpha(1-\alpha)^j X_{t-j}$$

递推公式: $\hat{X}_{T+1|T} = \quad \Leftrightarrow \text{ARIMA}(1, 1, 1)$

(4) 核光滑法 $\hat{T}_t = \sum_{i=1}^T a_i X_i$, 其中 $a_i = \frac{K(\frac{i-t}{h})}{\sum_{i=1}^T K(\frac{i-t}{h})}$

且: $K(\cdot)$ 称为核函数, 特征: 轴对称, 且中间高, 两边低.

通常取标准正态分布的密度函数.

(5) 局部加权多项式回归法 (LOWESS): 用加权最小二乘法拟合多项式.

2. 含季节性

(1). 差分法消除趋势和季节

$$\Delta X_t = X_t - X_{t-1} = (1-B)X_t$$

$$X_{t-1} \stackrel{\wedge}{=} BX_t$$

$$Y_t = \Delta X_t \Rightarrow X_t = X_{t-1} + Y_t$$

$$\Delta_s X_t = X_t - X_{t-s} = (1-B^s)X_t \quad (s \neq 1)$$

$$B: X_t \mapsto BX_t = X_{t-1}$$

$$\Rightarrow \Delta \Delta_s X_t \text{ 或 } \Delta_s \Delta X_t$$

(2) 时间序列模型 — 随机过程

— 列随机变量 $\{X_t, t=0, \pm 1, \pm 2, \dots\}$

第6周 (10月12日) 理论课

平稳时间序列模型 $\begin{cases} \text{平稳性} \\ \text{ACF} \end{cases}$

(1) 随机漫步 (Random Walk, RW) 非平稳. 并假设 $\{z_t\} \sim WN(0, \sigma_z^2)$

$$X_t = X_{t-1} + z_t \text{ (RW)} \longrightarrow X_t = z_1 + z_2 + \dots + z_t, \quad E(X_t) = 0, \quad \text{Var}(X_t) = t\sigma_z^2, \quad \nabla X_t = z_t$$

$$X_t = \mu + X_{t-1} + z_t \text{ (RW-drift)} \longrightarrow X_t = \mu t + z_1 + \dots + z_t, \quad E(X_t) = 0, \quad \text{Var}(X_t) = t\sigma_z^2, \quad \nabla X_t = \mu + z_t$$

补充: 白噪声序列 (white noise): 具有有限均值和有限方差的 i.i.d 随机变量序列 (平稳)

(2) 中心化: $E(X_t) = \mu$, 令 $Y_t = X_t - \mu$, 则 $E(Y_t) = 0$, 其中 $X_t = \varphi_1 X_{t-1} + z_t$

$$\text{Cov}(Y_t, Y_{t+k}) = E(Y_t Y_{t+k}) - 0 = E(X_t - \mu)(X_{t+k} - \mu) = \text{Cov}(X_t, X_{t+k})$$

(3) 线性过程:

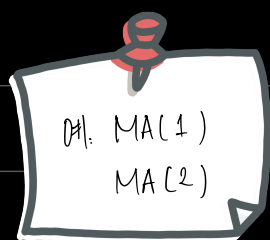
定理: 设 $\{Y_t\}$ 平稳, $\text{ACVF } \gamma_r(\cdot)$, $\{\psi_i\}$ 满足 $\sum |\psi_i| < +\infty$ (绝对求和收敛). 则 $X_t = \sum_{i=-\infty}^{+\infty} \psi_i Y_{t-i}$ 是平稳, 且 $\gamma_x(k) = \sum_i \sum_j \psi_i \psi_j \gamma_y(i+k-j)$

推论: $X_t = \sum_{i=-\infty}^{+\infty} \psi_i z_{t-i}$ 是平稳的, 称为线性过程. 且 $\gamma_x(k) = \sigma_z^2 \sum_{i=-\infty}^{+\infty} \psi_i \psi_{i+k}$ (因为: $\gamma_y(i+k-j) \neq 0 \Leftrightarrow i+k-j=0$, 即 $j=i+k$)

(4) MA(q)

$X_t = z_t + \theta_1 z_{t-1} + \dots + \theta_q z_{t-q}$, 此时 $\psi_i = \begin{cases} \theta_i, & 0 \leq i \leq q \text{ (有限项)} \\ 0, & \text{其他} \end{cases}$ 则 $\{X_t\}$ 平稳, 称为 q 阶移动平均过程.

$$\gamma_k = \sigma_z^2 \sum_{i=0}^{q-k} \psi_i \psi_{i+k} = \begin{cases} \sigma_z^2 \sum_{i=0}^{q-k} \theta_i \theta_{i+k}, & 0 \leq k \leq q \\ 0, & k > q \end{cases}, \quad \gamma(0) = \sigma_z^2 (1 + \theta_1^2 + \dots + \theta_q^2), \quad \text{ACF: } \rho_k = \frac{\gamma_k}{\gamma_0}, \quad \text{其中 } k > q \text{ 时, } \rho_k = 0 \quad (q \text{ 阶截尾})$$



$$\text{例: } MA(1) = z_t + \theta_1 z_{t-1}, \rho_k = \begin{cases} \frac{\theta_1^2}{1 + \theta_1^2}, & k=1 \\ 0, & k>1 \end{cases}$$

$$MA(2) = z_t + \theta_1 z_{t-1} + \theta_2 z_{t-2}, \rho_k = \begin{cases} \frac{\theta_1 + \theta_1 \theta_2}{1 + \theta_1^2 + \theta_2^2}, & k=1 \\ \frac{\theta_1}{1 + \theta_1^2 + \theta_2^2}, & k=2 \\ 0, & k>2 \end{cases}$$
$$\begin{cases} 1z_t + \theta_1 z_{t-1} + \theta_2 z_{t-2} \rightarrow X_t \\ z_{t+1} + \theta_1 z_t + \theta_2 z_{t-1} \rightarrow X_{t+1} \\ 1z_t + \theta_1 z_{t-1} + \theta_2 z_{t-2} \rightarrow X_t \\ z_{t+2} + \theta_1 z_{t+1} + \theta_2 z_t \rightarrow X_{t+2} \end{cases}$$

(五) 可逆性

$$A: X_t = z_t + \theta_1 z_{t-1}$$

$$B: X_t = z_t + \frac{1}{\theta_1} z_{t-1}$$

第7周 (10月19日) 理论课

$$\text{Review: } X_t = \mu + z_t + \theta_1 z_{t-1} + \dots + \theta_q z_{t-q}, MA(q) = \theta(B) z_t$$

$$\text{推广: } X_t = z_t + \varphi_1 z_{t-1} + \varphi_1^2 z_{t-2} + \varphi_1^3 z_{t-3} + \dots, |\varphi_1| < 1, MA(\infty) \Rightarrow X_t = \varphi X_{t-1} + z_t, AR(1)$$

$$(一) AR(1): X_t = \varphi X_{t-1} + z_t$$

$$* \text{线性组合 } X_t = \sum_{j=-\infty}^{\infty} \psi_j z_{t-j} \text{ 平稳.}$$

$$\textcircled{1} |\varphi_1| < 1, X_t \text{ 平稳}, X_t = z_t + \varphi_1 z_{t-1} + \varphi_1^2 z_{t-2} + \dots, \text{等比级数收敛, 又称作“因果平稳过程”}$$

* 因果平稳过程: X_t 可以表示当前或过去时刻扰动的收敛线性组合, 即 $MA(\infty)$ 形式, 此时称 X_t 是“因果 (Causal) 过程”。

$$\textcircled{2} |\varphi| = 1, X_t \text{ 非平稳}$$

$$\textcircled{3} |\varphi| > 1, X_t = -\frac{1}{\varphi_1} X_{t+1} - \frac{1}{\varphi_1^2} X_{t+2} - \frac{1}{\varphi_1^3} X_{t+3} - \dots \text{ 平稳}$$

故得: i. $AR(1)$ 平稳 $\Leftrightarrow |\varphi_1| \neq 1$

ii. $AR(1)$ 因果平稳 $\Leftrightarrow |\varphi_1| < 1$

$$\boxed{\text{证1}} \quad X_t = \varphi X_{t-1} + z_t$$

$$\Rightarrow X_t X_{t-k} = \varphi X_{t-1} X_{t-k} + z_t X_{t-k}$$

$$\Rightarrow \gamma(k) = \varphi_1 \gamma(k-1) + 0$$

$$\Rightarrow \rho(k) = \varphi_1 \cdot \rho(k-1), \forall k \geq 1.$$

$$\Rightarrow \rho(k) = \varphi_1^k, \forall k \geq 1$$

$$\boxed{\text{证2}} \quad (1 - \varphi B) X_t = z_t$$

$$\text{设 } X_t = \sum_{j=0}^{\infty} \pi_j z_{t-j} = \sum_{j=0}^{\infty} \pi_j B^j z_t$$

$$(1 - \varphi B) (\underbrace{\pi_0 + \pi_1 B + \pi_2 B^2 + \dots}_{\Rightarrow \pi(B)}) = 1$$

$$\pi_0 + [\pi_1 - \varphi_1 \pi_0] B + [\pi_2 - \varphi_1 \pi_1] B^2 + \dots = 1$$

$$\Rightarrow \pi_0 = 1, \pi_1 = \varphi_1, \pi_2 = \varphi_1^2, \dots$$

$$(一) AR(p): X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + \dots + \varphi_p X_{t-p} + z_t$$

$$\Rightarrow (1 - \varphi_1 B - \varphi_2 B^2 - \dots - \varphi_p B^p) X_t = z_t, \varphi(B) X_t = z_t$$

$$\pi(B) = 1 + \varphi_1 B + \varphi_1^2 B^2 + \dots, \text{称为 } (1 - \varphi B)^{-1}$$

$$X_t = \frac{1}{1 - \varphi_1 B} z_t = \sum_{j=0}^{\infty} (\varphi_1 B)^j z_t$$

$$(二) AR(2): X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + z_t$$

$$\Rightarrow (1 - \varphi_1 B - \varphi_2 B^2) X_t = Z_t \Rightarrow (1 - \pi_1 B)(1 - \pi_2 B) X_t = Z_t \Rightarrow \begin{cases} (1 - \pi_2 B) X_t = Y_t \\ (1 - \pi_1 B) Y_t = Z_t \end{cases}$$

* AR(2) 因果平稳 $\Leftrightarrow |\pi_1| < 1, |\pi_2| < 1$, 即 $\varphi(B) = 0$ 根在单位圆外.

* AR(p) 因果平稳 $\Leftrightarrow \varphi(B) = 0$ 根在单位圆外.

* $X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + Z_t, t \geq 1 \Rightarrow X_t X_{t-k} = \varphi_1 X_{t-1} X_{t-k} + \varphi_2 X_{t-2} X_{t-k} + Z_t X_{t-k}$

$$\Rightarrow \rho_k = \varphi_1 \rho_{k-1} + \varphi_2 \rho_{k-2}, \forall k \geq 1, \rho_0 = 1, \rho_1 = \varphi_1 \rho_0 + \varphi_2 \rho_{-1} \Rightarrow \rho_1 = \frac{\varphi_1}{1 - \varphi_2}$$

$$(1 - \varphi_1 B - \varphi_2 B^2) X_t = Z_t \Rightarrow \rho_k = A_1 \pi_1^k + A_2 \pi_2^k, \text{ 其中 } \pi_1, \pi_2 \text{ 是方程 } y^2 - \varphi_1 y - \varphi_2 = 0 \text{ 根, 特征根.}$$

故: 因果平稳 $\Leftrightarrow |\pi_1| < 1, |\pi_2| < 1, \rho_k \rightarrow 0$, 当 $k \rightarrow +\infty$.

例: $X_t = \frac{1}{3} X_{t-1} + \frac{2}{7} X_{t-2} + Z_t$

解: 解方程 $y^2 - \frac{1}{3}y - \frac{2}{7} = 0$ 得 $y = \frac{2}{7}$ 或 $y = -\frac{1}{3} \Rightarrow \text{ACF: } \rho(k) = \frac{16}{21} (\frac{2}{7})^k + \frac{5}{21} (-\frac{1}{3})^k$

* $(1 - \varphi_1 B - \varphi_2 B^2) = (1 - \pi_1 B)(1 - \pi_2 B)$

Vieta 定理: $\begin{cases} \pi_1 \pi_2 = -\varphi_2 \\ \pi_1 + \pi_2 = \varphi_1 \end{cases}$

第8周 (10A26A) 理论课

ARMA(p, q): $X_t = \varphi_1 X_{t-1} + \dots + \varphi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q} \Rightarrow \varphi(B) X_t = \theta(B) Z_t$

* $X_t = \theta(B) Z_t \xrightarrow{\text{可逆}} Z_t = \frac{1}{\theta(B)} X_t = \left(\sum_{j=0}^{\infty} \pi_j B^j \right) X_t$

* $\varphi(B) X_t = Z_t \xrightarrow{\text{因果}} X_t = \frac{1}{\varphi(B)} Z_t = \left(\sum_{j=0}^{\infty} \pi_j B^j \right) Z_t$

$\Rightarrow \text{ARMA}(p, q) \text{ 因果} \Leftrightarrow \varphi(B) = 0 \text{ 根在单位圆外}, X_t = \frac{\theta(B)}{\varphi(B)} Z_t = \left(\sum_{j=0}^{\infty} \pi_j B^j \right) \theta(B) Z_t \text{ 可逆} \Leftrightarrow \theta(B) = 0 \text{ 均根在单位圆外}$

例: $X_t = \varphi_0 + \varphi_1 X_{t-1} + Z_t + \theta_1 Z_{t-1}$

解: 因果平稳 $\Leftrightarrow |\varphi_1| < 1, |\theta_1| < 1$.

设 $E(X_t) = \mu, \mu = \varphi_0 + \varphi_1 \mu \Rightarrow \mu = \frac{\varphi_0}{1 - \varphi_1}$

$\Rightarrow X_t - \mu = \varphi_1 (X_{t-1} - \mu) + Z_t + \theta_1 Z_{t-1}, \varphi(B) = 1 - \varphi_1 B, \theta(B) = 1 + \theta_1 B$

不妨设 $\varphi_0 = 0, \text{Var}(X_t) = \sigma^2, \sigma^2 = \varphi_1^2 \sigma^2 + \sigma_z^2 + \theta_1^2 \sigma_z^2 + 2\theta_1 \varphi_1 E(X_{t-1} Z_{t-1}) \stackrel{= \sigma_z^2}{=} \Rightarrow \sigma^2 = \frac{1 + \theta_1^2 + 2\theta_1 \varphi_1}{1 - \varphi_1^2} \sigma_z^2$

③ ACF: $X_t = \varphi_1 X_{t-1} + Z_t + \theta_1 Z_{t-1}$

$\Rightarrow X_t X_{t-k} = \varphi_1 X_{t-1} X_{t-k} + Z_t X_{t-k} + \theta_1 Z_{t-1} X_{t-k} \Rightarrow \gamma_k = \varphi_1 \gamma_{k-1} + 0 + \begin{cases} 0, & k > 1 \\ \theta_1 \sigma_z^2, & k = 1 \end{cases}$

$$k=1 \text{ 时}, \rho_1 = \phi_1 + \frac{\theta_1 \sigma_z^2}{\gamma_0}$$

$$k > 1 \text{ 时}, \rho_k = \phi_1 \rho_{k-1} \rightarrow 0 \quad (k \rightarrow \infty) \quad (\text{拖尾性})$$

(-) ACF: $\rho_k = \text{Corr}(X_t, X_{t-k})$, 对 MA(q), $\rho_k = 0$, 当 $k > q$ 时. 偏自相关系数 PACF (Partial ACF): $\text{Corr}(X_t, X_{t-2} | X_{t-1}) \triangleq \phi_{22}$

一般地 $\text{Corr}(X_t, X_{t-k} | X_{t-1}, X_{t-2}, \dots, X_{t-k+1}) \triangleq \phi_{kk} \triangleq \text{Corr}(X_t - \hat{X}_t, X_{t-k} - \hat{X}_{t-k})$, 其中 $\hat{X}_t = E(X_t | X_{t-1}, \dots, X_{t-k+1})$

对 AR(p) 来讲, $\phi_{kk} = 0$, 当 $k > p$ 时, $\hat{X}_t = E(\phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + z_t | X_{t-1}, \dots, X_{t-k+1}) = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p}$.

$$\text{于是 } X_t - \hat{X}_t = z_t$$

(=) PACF 定义: $X_t \sim X_{t-1} + \dots + X_{t-k+1} + X_{t-k}$, $X_t = \phi_{k1} X_{t-1} + \phi_{k2} X_{t-2} + \dots + \phi_{kk} X_{t-k} + \rho_t$

$$X_{t-k} \text{ 前的回归系数即 } \phi_{kk}. \min E(X_t - \phi_{k1} X_{t-1} - \dots - \phi_{kk} X_{t-k})^2$$

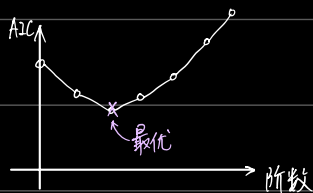
第9周 (11月2日) 理论课

ARMA / ARIMA: (1) 定阶; (2) 估计; (3) 检验; (4) 优化.

(-) 定阶:

1. ACF / PACF: ACF: q 阶截尾 $\rightarrow$ MA(q)

PACF: p 阶截尾 $\rightarrow$ AR(p)



2. AIC / BIC: AIC (Akaike Information Criterion 赤池信息准则) $= -2 \ln L(\hat{\theta}) + 2 \frac{\text{参数个数} + 1}{n} \propto \text{RSS (残差平方和)}$

$$\text{AICc} = -2 \ln L(\hat{\theta}) + \left[2(p+q+1) \frac{n}{n-p-q-2} \right] \propto T \text{ (样本量)} \quad (\text{使用定阶})$$

$\Rightarrow$ 推荐: BIC (Bayes) $= -2 \ln L(\hat{\theta}) + \left[\ln(n) \cdot (p+q+1) \right]$ 惩罚力度比 AIC 大 (使用低阶) 又被称作 SIC (Schwarz)

(=) 估计: 条件最小二乘法 (CSS) & 极大似然估计 (MLE)

$$X_t - \mu = \phi_1 (X_{t-1} - \mu) + \phi_2 (X_{t-2} - \mu) + \phi_3 (X_{t-3} - \mu) + z_t \Rightarrow \phi(B)(X_t - \mu) = \theta(B)z_t, z_t \sim WN(0, \sigma_z^2)$$

$$1. \hat{\mu} = \bar{X}$$

$$2. \text{CSS: } X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + z_t, \min S_c = \sum_{t=p+1}^T (X_t - \phi_1 X_{t-1} - \dots - \phi_p X_{t-p})^2$$

$$\text{ARMA}(p, q): \min S_c = \sum (X_t - \phi_1 X_{t-1} - \dots - \phi_p X_{t-p} + \theta_1 z_{t-1} + \dots + \theta_q z_t)^2$$

$$3. \text{MLE: } f(x_1, x_2, \dots, x_T, \theta) = f(x_1) f(x_2 | x_1) f(x_3 | x_1, x_2) \dots f(x_T | x_1, x_2, \dots, x_{T-1}) \leftarrow f(x_1, x_2) = f(x_1) f(x_2) = f(x_2 | x_1) \cdot f(x_1)$$

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + z_t + \theta_1 z_{t-1} + \dots + \theta_q z_{t-q} \Rightarrow (X_t | X_{t-1}, X_{t-2}, \dots) \sim N(\phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \theta_1 z_{t-1} + \theta_q z_{t-q}, \sigma_z^2)$$

第10周理论课(11月9日)

(一) 预测

$X_1, X_2, \dots, X_n$ 已知: n 预测原点. h 步长. $\mathcal{F}_n$ (Filter): 到 n 时刻为止的历史信息.

用 $f(\mathcal{F}_n)$ 预测 X_{n+h} : $X_{n+h} - f(\mathcal{F}_n)$

$$\min_f (X_{n+h} - f(\mathcal{F}_n))^2 = E \left[X_{n+h} - E(X_{n+h} | \mathcal{F}_n) + E(X_{n+h} | \mathcal{F}_n) - f(\mathcal{F}_n) \right]^2$$

$$= E \left(X_{n+h} - E(X_{n+h} | \mathcal{F}_n) \right)^2 + E \left(E(X_{n+h} | \mathcal{F}_n) - f(\mathcal{F}_n) \right)^2$$

当且仅当 $f(\mathcal{F}_n) = E(X_{n+h} | \mathcal{F}_n)$ 时, $E(X) = E(E(X|Y)) = E[X_{n+h} - E(X_{n+h} | \mathcal{F}_n) | \mathcal{F}_n]$

$= E(X_{n+h} | \mathcal{F}_n) - E(X_{n+h} | \mathcal{F}_n) = 0$. 回归分析 $Y = E(Y|X) + u$

$$\Rightarrow f(\mathcal{F}_n) = E(X_{n+h} | \mathcal{F}_n) \quad \text{即} \quad \arg \min_f E(X_{n+h} - f(\mathcal{F}_n))^2 = E(X_{n+h} | \mathcal{F}_n)$$

即最优的方误差预测为预测目标在给定历史信息下的条件期望.

$$X_t, Y_t = X_t - \mu, \quad \hat{X}_n(h) = \hat{Y}_n(h) + \mu.$$

(二) AR(1): $X_t = \phi X_{t-1} + Z_t$

$$\text{一步预测: } \hat{X}_n(1) = E(X_{n+1} | \mathcal{F}_n) = E(\phi X_n + Z_{n+1} | \mathcal{F}_n) = \phi X_n + E(Z_{n+1}) = \phi X_n$$

$$\text{两步预测: } \hat{X}_n(2) = E(X_{n+2} | \mathcal{F}_n) = E(\phi X_{n+1} + Z_{n+2} | \mathcal{F}_n) = \phi E(X_{n+1} | \mathcal{F}_n) = \phi \hat{X}_n(1) = \phi^2 X_n$$

$\vdots$

$$\hat{X}_n(h) = \phi^h X_n \rightarrow 0 = E(X_t) \quad (\text{金融上: 均值反转}) \quad \text{半衰期: } \hat{X}_n(h) = \frac{1}{2} X_n \Rightarrow \text{求 } h.$$

均值回转 (均值反转) (mean reversion): 当预测步长趋于无穷大时, 预测收敛于序列均值. $\left. \begin{array}{l} \text{均值回转速度用半衰期 (half life) 来衡量: Def: } l = \frac{\ln(0.5)}{\ln(1/\phi_1)} \text{, 即满足 } \hat{X}_n(l) = \frac{1}{2} X_n \end{array} \right\} \text{平稳时间序列特征.}$

$$\text{均值回转速度用半衰期 (half life) 来衡量: Def: } l = \frac{\ln(0.5)}{\ln(1/\phi_1)} \text{, 即满足 } \hat{X}_n(l) = \frac{1}{2} X_n$$

$$e_n(1) = X_{n+1} - \hat{X}_n(1) = Z_{n+1},$$

$$e_n(2) = X_{n+2} - \hat{X}_n(2) = \phi X_{n+1} + Z_{n+2} - \phi \hat{X}_n(1) = Z_{n+2} + \phi Z_{n+1}$$

$\vdots$

$$e_n(h) = Z_{n+h} + \phi Z_{n+h-1} + \dots + \phi^{h-1} Z_{n+1}$$

$$\text{则 } \text{Var}(e_n(h)) = (1 + \phi_1^2 + \dots + \phi_1^{2(h-1)}) \sigma_z^2, \text{ 由于 } E(e_n(h)) = 0, \text{ 则 } \frac{e_n(h)}{\text{Var}(e_n(h))}$$

(三) MA(1): $X_t = \mu + Z_t + \theta Z_{t-1}$

$$\text{一步预测: } \hat{X}_n(1) = E(\mu + Z_{n+1} + \theta Z_n | \mathcal{F}_n) = \mu + \theta Z_n \quad e_n(1) = Z_{n+1}$$

两步预测: $\hat{X}_n(2) = E(\mu + z_{n+2} + \theta_1 z_{n+1} | \mathcal{F}_n) = \mu$ (均值回归)

$$ARMA(1,1): X_t = \phi_1 X_{t-1} + z_t + \theta_1 z_{t-1}$$

$$\hat{X}_n(1) = E(\phi_1 X_n + z_{n+1} + \theta_1 z_n) = \phi_1 X_n + \theta_1 z_n$$

$$\hat{X}_n(2) = E(\phi_1 X_{n+1} + z_{n+2} + \theta_1 z_{n+1}) = \phi_1 \hat{X}_n(1)$$

$$ARIMA(1,1,1): (1-\phi_1 B)(1-B)X_t = z_t + \theta_1 z_{t-1} \Rightarrow X_t = (1+\phi_1)X_{t-1} - \phi_1 X_{t-2} + z_t + \theta_1 z_{t-1}$$

$$\hat{X}_n(1) = E_n(X_n + \phi_1 X_n + \phi_1 X_{n-1} + z_{n+1} + \theta_1 z_{n+1}) = E(\phi_1 X_n + z_{n+1} | \mathcal{F}_n) = \phi_1 X_n + E(z_{n+1}) = \phi_1 X_n$$

$$\hat{X}_n(1) = E_n(X_{n+1}) = E_n(\hat{\phi}_1 X_{n+1} + z_{n+1}) = \hat{\phi}_1 X_{n+1}$$

$$\phi_1 X_n + z_n - \hat{\phi}_1 X_n = (\phi_1 - \hat{\phi}_1) X_n + z_n$$

$$RW: \nabla X_t = z_t \Rightarrow X_t - X_{t-1} = z_t \Rightarrow X_t = X_{t-1} + z_t \quad \text{股票} \nabla X_t \text{ 代表收益率, 而 } \nabla X_t \sim WN(0, \sigma^2)$$

$$\hat{X}_n(h) = X_n$$

非要预测的话, 下一天的预测值只能为当天实际值.

第11周 11月16日 理论课

(一) 回溯检验 Backtesting / 样本外



$$\text{for } j = h, h+1, \dots, T, \{X_t, t \leq j\} \Rightarrow M \Rightarrow \hat{x}_j(1), e_j(1) = x_{j+1} - \hat{x}_j(1)$$

$$\frac{\sum [e_j(1)]^2}{T-h} \text{ 或 } \frac{\sum |e_j(1)|}{T-h}$$

(二) AIC/BIC 样本估计

(三) 季节模型

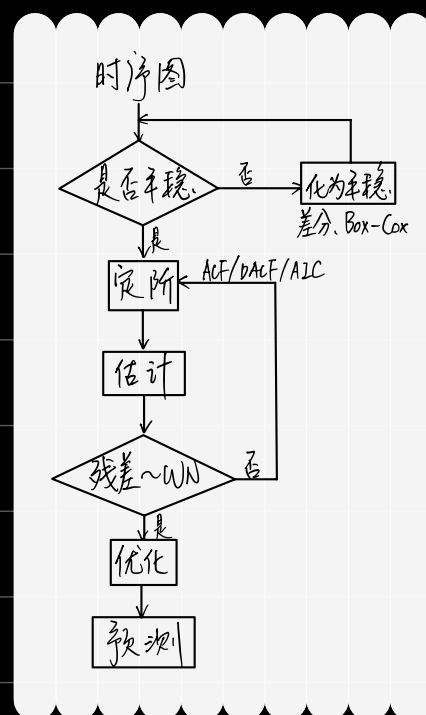
$$\text{无季节: } \phi(B)(1-B)f(X_t) = \theta(B)z_t$$

$$\text{有季节: } (1-B^s)(1-B)X_t \sim ARIMA$$

$$\nabla_s X_t = X_t - X_{t-s} = (1-B^s)X_t$$

$$\text{乘积季节模型: } X_t = (1+\theta B)(1+\theta B^2)z_t = (1+\theta B + \theta B^2 + \theta \theta B^3)z_t$$

平稳: ARMA, 不平稳: ARIMA.



第12周 (11月23日) 理论课.

单位根检验 (Unitroot Test)

问题 1: $X_t = \phi_1 X_{t-1} + Z_t$, $H_0: \phi_1 = 1 \iff H_1: \phi_1 < 1$ (RW) (AR(1)) $H_0: X_t = X_{t-1} + Z_t \Rightarrow \nabla X_t = Z_t$ (RW)

问题 2: $X_t = \phi_0 + \phi_1 X_{t-1} + Z_t$, $H_0: \phi_1 = 1 \iff H_1: \phi_1 < 1$ (RW-d) (AR(1)) $H_0: X_t = \phi_0 + X_{t-1} + Z_t \Rightarrow \nabla X_t = \phi_0 + Z_t$ RW的漂移项 (drift)

问题 3: $X_t = \beta_0 + \beta_1 t + V_t$ Differential Stationary 差分平稳 $V_t = \phi_1 X_{t-1} + Z_t \Rightarrow (1 - \phi_1 B)(\nabla X_t - \beta_1) = Z_t$

Trend Stationary 趋势平稳

TS: $X_t = \beta_0 + \beta_1 t + V_t$ V_t ~ ARMA

$$\Rightarrow X_t - \beta_0 - \beta_1 t = \phi_1 [X_{t-1} - \beta_0 - \beta_1(t-1)] + Z_t$$

$$\Rightarrow X_t = \beta_0(1 - \phi_1) + \phi_1 \beta_0 + \beta_1(1 - \phi_1)t + \phi_1 X_{t-1} + Z_t$$

$$= \omega(\text{常数}) + \delta t + \phi_1 X_{t-1} + Z_t$$

从而得到回归模型: $X_t = \omega + \delta t + \phi_1 X_{t-1} + Z_t$ (DS) (TS) $H_0: \phi_1 = 1 \iff H_1: \phi_1 < 1$

一、进行检验 (以问题 1 为例) — Dickey-Fuller 检验.

在备择假设 $H_1: \phi_1 < 1$ 时, 即 X_t 平稳的情况下 $\frac{\hat{\phi}_1 - \phi_1}{\sqrt{\text{Var}(\hat{\phi}_1)}} \sim N(0, 1)$, 但在原假设下不成立.

于是构造统计量 $\tau = \frac{\hat{\phi}_1 - \phi_1}{\sqrt{\text{Var}(\hat{\phi}_1)}} \sim \frac{\frac{1}{2}(X_1^2 - 1)}{[\int_0^1 W(t)^2 dt]^{\frac{1}{2}}}$, 其中 $W(t)$ 布朗运动 $\mu=0, \sigma^2=t$

* 推广: $X_t = \phi_1 X_{t-1} + Z_t \iff \nabla X_t = (\phi_1 - 1)X_{t-1} + Z_t = \gamma X_{t-1} + Z_t$, $H_0: \gamma = 1 \iff H_1: \gamma < 1$. $\tau = \frac{\hat{\gamma} - \gamma}{\sqrt{\text{Var}(\hat{\gamma})}}$

AR(p) 变形为 $\nabla X_t = \gamma X_{t-1} + \sum_{j=2}^p \beta_j \nabla X_{t-j} + Z_t = \gamma X_{t-1} + \sum_{j=1}^{p-1} \beta_j \nabla X_{t-j} + Z_t$

以 AR(2) 为例: 问题 1: $\nabla X_t = \gamma X_{t-1} + \beta_1 \nabla X_{t-1} + Z_t$ ("none") tan1 $H_0: \gamma = 0 \iff H_1: \gamma < 0$

问题 2: $\nabla X_t = \phi_0 + \gamma X_{t-1} + \beta_1 \nabla X_{t-1} + Z_t$ ("drift") tan2 $H_0: \phi_0 = \gamma = 0$

问题 3: $\nabla X_t = \phi_0 + \delta t + \gamma X_{t-1} + \beta_1 \nabla X_{t-1} + Z_t$ ("trend") tan3 $H_0: \phi_0 = \delta = \gamma = 0$ phi1 phi2 phi3

第13周 (11月30日) 理论课.

(-) ARMAX: $y_t = \beta_0 + \beta_1 x_t + e_t$

CLM: $e_t \sim WN(0, \sigma^2) \Rightarrow$ 拟合 (最小二乘) $\min \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_t)^2$

经典线性模型

e_t 平稳 ($e_t \sim ARMA$) {x_t} 和 {y_t} 平稳

$y_t - \beta_0 - \beta_1 x_t$ 平稳 $\iff \{x_t\}$ 和 $\{y_t\}$ 具有协整关系

是否平稳		回归方程
x_t	y_t	
✓	✓	$y_t \sim x_t$
x	✓	$\nabla y_t \sim x_t$
x	x	e_t 平稳: $y_t \sim x_t$
		$\nabla y_t \sim \nabla x_t$

若 e_t 不平稳, 则回归模型不成立. (即使有回归关系也不成立, 称为虚假回归)

* 若有 $y_t - \beta_0 - \beta_1 x_t \sim \text{ARMA}$, 则称 $y_t \sim \text{ARMAX}$.

$$\phi(B)(y_t - \beta_0 - \beta_1 x_t) = \theta(B)z_t \Leftrightarrow y_t = \beta_0 + \beta_1 x_t + e_t, \quad \phi(B)e_t = \theta(B)z_t.$$

④ 协整: 由于时间序列模型的滥用, 由 Granger 提出协整理论及方法, 即残差 (e_t) 平稳.

(一) ARFIMA (3 种)

长记忆 (Long Memory): $(1-B)^d X_t = z_t, \quad -\frac{1}{2} < d < \frac{1}{2}, \quad \rho_k \approx c \cdot k^{2d-1}.$

$$X_t = \mu_t + z_t, \quad \mu = E(X_t | \mathcal{F}_{t-1})$$

? $\text{Var}(X_t | \mathcal{F}_{t-1}) = \text{Var}(z_t | \mathcal{F}_{t-1}) = E(z^2 | \mathcal{F}_{t-1})$ 随 t 变化.

(二) 回报率 (Return) $\neq$ 状态.

$r_t = \ln(1 + R_t)$ 对数回报. 其中, 考虑区间 $(t-k, t]$, 收益率 $R_t[k] = \frac{P_t - P_{t-k}}{P_{t-k}} \times 100\%$.

第14周 (12月9日) 实验课

(一) 递推: $R_t = R_t[1] = \frac{P_t - P_{t-1}}{P_{t-1}} \Leftrightarrow P_t = P_{t-1}(1 + R_t)$

$r_t = \ln(1 + R_t) = \nabla \ln P_t \Leftrightarrow P_t = P_{t-1} \cdot e^{r_t}$ 由于 $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$, 于是 $r_t = \ln(1 + R_t) \approx R_t$.

(二) 特征

1. 偏度 (Skewness): 标准化的三阶中心矩, $S(x) = \frac{E(X-\mu)^3}{\sigma^3}$. 刻画分布的对称性, 对称分布 $S(x) = 0$.

2. 峰度 (Kurtosis): 标准化的四阶中心矩. $K(x) = \frac{E(X-\mu)^4}{\sigma^4}$. 刻画分布的尾部特征.

* 正态分布 $K(x) = 3$, 于是将 $K(x) - 3$ 称为超额峰度, 具有正的超额峰度 ($K(x) > 3$) 的分布称为 尖峰厚尾分布. (heavy tail)

• 分布检验: • 检验对称性: $H_0: S(x) = 0 \Leftrightarrow H_1: S(x) > 0$. $t_s = \frac{\hat{S}(x)}{\sqrt{6/T}} \sim N(0, 1)$

• 检验厚尾性: $H_0: K(x) = 3 \Leftrightarrow H_1: K(x) > 3$, $t_k = \frac{\hat{K}(x) - 3}{\sqrt{24/T}} \sim N(0, 1)$

• 检验正态性: H_0 : 服从 $\Leftrightarrow H_1$: 不服从. Jarque-Bera 雅克-贝拉检验. $JB = \frac{\hat{S}(x)^2}{6/T} + \frac{(\hat{K}(x) - 3)^2}{24/T}$

* 分布假设检验 2 种 Quantile - Quantile QQ图. (Sample quantiles $\sim$ Theoretical quantiles) $JB = t_s^2 + t_k^2$

第15周 (12月14日) 理论课 · 条件异方差建模

一. 波动率 (Volatility)

$\{r_t\}$: 回报率: $r_t = \phi_0 + \phi_1 r_{t-1} + \dots + \phi_p r_{t-p} + \theta_1 z_{t-1} + \dots + \theta_q z_{t-q} + z_t = \mu_t + z_t$, $\mu_t = E[r_t | \mathcal{F}_{t-1}]$

$$\text{Var}[z_t | \mathcal{F}_{t-1}] = \text{Var}[z_t | \mathcal{F}_{t-1}] \triangleq \sigma_t^2 \quad (\text{Volatility: } \sigma_t = \sqrt{\text{Var}[r_t | \mathcal{F}_{t-1}]})$$

⇒ 基于此, 如果 $z \sim \text{Normal}(WN)$, 则 $\text{Var}[z_t | \mathcal{F}_{t-1}] = \text{Var}[z_t] = \sigma_z^2$, 但 $\{r_t\}$ 具有尖峰厚尾性, 与前者矛盾.

二. Volatility 经验特征

三. ARCH 效应 Test (条件异方差)

* 步骤:

1. 拟合均值方程 (ARMA 模型) 来消除线性依赖;
2. 对均值方程的残差进行 ARCH 效应 (条件异方差) 检验;
3. 若存在异方差, 则指定一个波动率模型并联合估计均值方程和方差方程.
4. 检验模型的充分性 (最终的标准残差应既无自相关性, 也无异方差)

$\sigma^2 \neq \text{常数}, \sigma^2 = E[z_t | \mathcal{F}_{t-1}]$
 $\mu_t = \text{常数}$
 $\Leftrightarrow r_t \sim WN \Leftrightarrow \{r_t\}$ 无自相关.
 $\rightarrow \sigma_t^2$ 条件异方差 $\Leftrightarrow \{z_t\}$ 有自相关.

① $\{\hat{z}_t^2\}$. ACF 或 Ljung-Box Test.

② Engle's ARCH Test: $\hat{z}_t^2 = \alpha_0 + \alpha_1 \hat{z}_{t-1}^2 + \dots + \alpha_m \hat{z}_{t-m}^2 + \varepsilon_t$. 检验: $H_0: \alpha_1 = \alpha_2 = \dots = \alpha_m = 0$

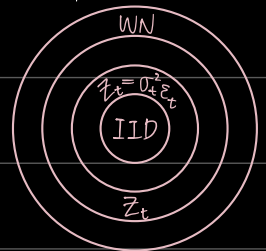
四. ARCH: Autoregressive conditional heteroscedasticity 自回归条件异方差.

Def. $z_t = \sigma_t \varepsilon_t$, 其中 $\varepsilon_t \stackrel{iid}{\sim} (0, 1)$, $\sigma_t^2 = \omega + \alpha_1 z_{t-1}^2 + \dots + \alpha_m z_{t-m}^2$, 称为 m 阶的自回归条件异方差模型. 记 ARCH(m)

$$* \text{Var}\left(\frac{z_t}{\sigma_t} | \mathcal{F}_{t-1}\right) = \frac{\text{Var}(z_t | \mathcal{F}_{t-1})}{\sigma_t^2} = \frac{\sigma_t^2}{\sigma_t^2} = 1, \text{ 其中 } \varepsilon_t = \frac{z_t}{\sigma_t}.$$

$$\Leftrightarrow z_t | \mathcal{F}_{t-1} \sim F(0, \sigma_t^2), r_t | \mathcal{F}_{t-1} \sim F(\mu_t, \sigma_t^2)$$

* 在有历史信息流情况下, $r_t = \mu_t + \sigma_t \varepsilon_t | \mathcal{F}_{t-1}$, μ_t, σ_t 为常数, 则 r_t 随 ε_t 分布.



* 例: ARCH(1)

$$\begin{cases} z_t = \sigma_t \varepsilon_t, \varepsilon_t \stackrel{iid}{\sim} N(0, 1) \\ \sigma_t^2 = \omega + \alpha_1 z_{t-1}^2 \end{cases}$$

性质

$$\textcircled{1} E[z_t] = E[\sigma_t \varepsilon_t] = E[E(\sigma_t \varepsilon_t | \mathcal{F}_{t-1})] = E[\sigma_t E(\varepsilon_t | \mathcal{F}_{t-1})] = 0;$$

$$\textcircled{2} E[z_t^2] = E[\sigma_t^2 \varepsilon_t^2] = E[E(\sigma_t^2 \varepsilon_t^2 | \mathcal{F}_{t-1})] = E[\sigma_t^2 E(\varepsilon_t^2 | \mathcal{F}_{t-1})] = E[\omega + \alpha_1 z_{t-1}^2]$$

$$\Rightarrow E[z_t^2] = \frac{\omega}{1-\alpha_1}, \quad 0 < \alpha_1 < 1$$

$$\textcircled{2} \text{Cov}(z_t, z_s) = E(z_t z_s) \neq 0$$

$$\textcircled{4} E[z_t^4] = E[E(\sigma_t^4 \varepsilon_t^4 | \mathcal{F}_{t-1})] = E[\sigma_t^4 E(\varepsilon_t^4 | \mathcal{F}_{t-1})] = 3 E[(\omega + \alpha_1 z_{t-1}^2)^2]$$

$$= 3 E[\omega^2 + \alpha_1^2 z_{t-1}^4 + 2\omega\alpha_1 z_{t-1}^2]$$

$$\rho = \frac{E(x)^4}{\sigma^4} = 3 \text{ (Normal)}$$

$$= \dots > 3$$

z_t 不是正态

第16周 12月21日 理论课 · 条件异方差建模

Review:

$$\begin{cases} r_t = \mu_t + z_t, \mu_t = E_{t-1}(r_t) & \text{AR模型} \\ z_t = \sigma_t \varepsilon_t, \varepsilon_t \stackrel{i.i.d.}{\sim} F(0,1) \\ \sigma_t^2 = \omega + \sum_{i=1}^m \alpha_i z_{t-i}^2 & \text{ARCH模型} \end{cases}$$

$$\text{令 } \eta_t = z_t^2 - \sigma_t^2, \text{ 鞅差序列, } E(\eta_t) = 0, \text{ 无相关性, 似WN.}$$

$$\text{ARCH}(m) \Leftrightarrow \{z_t^2\} \text{AR}(m), \hat{\varepsilon}_t = \frac{\hat{z}_t}{\hat{\sigma}_t};$$

$$\text{模型充分} \Leftrightarrow \{\varepsilon_t\} \stackrel{i.i.d.}{\sim} F(0,1)$$

* 从近似来讲, AR, MA, ARMA 均等价于 AR 模型, 但代价为阶数很高

$$\text{修正: } \text{AR}(1) \Rightarrow \text{MA}(\infty), \text{MA}(1) \Rightarrow \text{AR}(\infty), \text{else } \text{ARMA}(1,1) \Rightarrow \text{MA}(\infty) \Rightarrow \text{AR}(\infty)$$

一. GARCH 模型

GARCH(m,s) 假设: sstd 分布 (非对称)

$$\begin{cases} r_t = \mu_t + z_t, \mu_t = E_{t-1}(r_t) & \text{AR模型} \\ z_t = \sigma_t \varepsilon_t, \varepsilon_t \stackrel{i.i.d.}{\sim} F(0,1) \\ \sigma_t^2 = \omega + \sum_{i=1}^m \alpha_i z_{t-i}^2 + \sum_{i=1}^s \beta_i \sigma_{t-i}^2 & \text{GARCH模型} \end{cases}$$

$$* \text{GARCH}(m,s) \Leftrightarrow \{z_t^2\} \text{ARMA} \quad \eta_t = z_t^2 - \sigma_t^2$$

$$z_t^2 - \eta_t = \omega + \alpha_1 z_{t-1}^2 + \dots + \alpha_m z_{t-m}^2 + \beta_1 (z_{t-1}^2 - \eta_{t-1}) + \dots + \beta_s (z_{t-s}^2 - \eta_{t-s})$$

$$\Rightarrow z_t^2 = \omega + (\alpha_1 + \beta_1) z_{t-1}^2 + \dots + (\alpha_p + \beta_p) z_{t-p}^2 + \eta_t - \beta_1 \eta_{t-1} - \dots - \beta_s \eta_{t-s}, \text{ 其中 } p = m \vee s = \max(m,s)$$

即关于 $\{z_t^2\}$ 的 $\text{ARMA}(\max(m,s), s)$

则有 $GARCH(2,2) \Leftrightarrow \{z_t^2\} ARMA(2,2)$; $GARCH(1,2) \Leftrightarrow \{z_t^2\} ARMA(2,2)$; $GARCH(2,1) \Leftrightarrow \{z_t^2\} ARMA(2,1)$;

例. $GARCH(1,1) \Leftrightarrow ARMA(1,1)$

$$\begin{cases} \sigma_t^2 = \omega + \alpha_1 z_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ z_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0,1) \end{cases} \quad \begin{aligned} E[z_t^2] &= E[\sigma_t^2 \varepsilon_t^2] = E[E[(\omega + \alpha_1 z_{t-1}^2 + \beta_1 \sigma_{t-1}^2) \varepsilon_t^2] | I_t]] \\ &= E[\omega + \alpha_1 z_{t-1}^2 + \beta_1 \sigma_{t-1}^2 | I_t]] = E[\sigma_t^2] \end{aligned}$$

$\Rightarrow \{z_t\}$ 性质: $E[z_t] = 0$

$ARMA(1,1): X_t = \phi_0 + \phi_1 X_{t-1} + z_t + \theta_1 z_{t-1}$

无条件均值:

$$E[z_t^2] = \frac{\beta_1}{1 - \alpha_1 - \beta_1}$$

$$\Rightarrow E[X_t] = \frac{\phi_0}{1 - \phi_1}$$

$$K = \frac{3[1 - (\alpha_1 + \beta_1)^2]}{1 - (\alpha_1 + \beta_1)^2 - 2\alpha_1^2} = \frac{E[z_t^4]}{E[z_t^2]^2}$$

由上述讨论可得: $\sigma_t^2 = E[z_t^2 | I_{t-1}]$, $\hat{\sigma}_t$: 波动率, $\eta_t = z_t^2 - E[z_t^2 | I_{t-1}]$

$$[1] IGARCH: \sigma_t^2 = \omega + (1 - \beta_1) z_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

$\Rightarrow$ VaR: Value and Risk 模型. $\Rightarrow$ Risk Matrices. $Var[h] = \sqrt{h} Var[1]$. 根据当天 VaR, 预测 h 天后 VaR.

$$[2] TGARCH 模型 / GJR 模型: \sigma_t^2 = \omega + \sum_{i=1}^m (\alpha_i + \gamma_i \underbrace{I(z_{t-i} < 0)}_{\text{Sig 符号函数}}) z_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2 \quad \text{华人统计学家 H. Tong.}$$

门槛 Threshold

Sig 符号函数.

$\downarrow$
不对称